

Kummer-Fourier Collision Variance and Weighted Hyperplane Matroid Sums

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Abstract

This note corrects a tempting but false linear product formula for Kummer collision excess. The corrected identity is exact finite algebra: for a CRT product of local Kummer obstruction indicators, the collision variance of the induced obstruction-vector distribution is a Plancherel sum whose local factors are squared complex characteristic functions. Equivalently, the variance is a weighted hyperplane-arrangement sum over the dual of the Kummer log matroid. The formula is unconditional and finite; no analytic equidistribution theorem, PHNC input, or Erdős–Graham resolution is asserted.

1 Setup

Let $\mathcal{Q}$ be a finite set of primes and put

$$P = \prod_{q \in \mathcal{Q}} q.$$

Let ℓ be prime, and let $x_q \in \mathbb{F}_\ell^r$ be the Kummer log row attached to q for a fixed rank- r base vector $\mathbf{a} = (a_1, \dots, a_r)$. Write

$$\Gamma_q = \langle a_1, \dots, a_r \rangle \subset \mathbb{F}_q^*$$

and

$$p_q = \frac{|\Gamma_q|}{q-1}.$$

For $m \in (\mathbb{Z}/P\mathbb{Z})^*$, define the local obstruction indicator

$$B_q(m) = \mathbf{1}_{\Gamma_q}(-m_q^{-1}),$$

where m_q denotes the image of m in $\mathbb{F}_q^*$. By the Chinese remainder theorem, as m varies uniformly over $(\mathbb{Z}/P\mathbb{Z})^*$, the variables B_q are independent Bernoulli variables with

$$\mathbb{P}(B_q = 1) = p_q.$$

Definition 1 (Bernoulli-Kummer obstruction vector). *Define*

$$V(m) = \sum_{q \in \mathcal{Q}} B_q(m) x_q \in \mathbb{F}_\ell^r.$$

Let

$$N(v) = \#\{m \in (\mathbb{Z}/P\mathbb{Z})^* : V(m) = v\}, \quad v \in \mathbb{F}_\ell^r,$$

and

$$\bar{N} = \frac{\varphi(P)}{\ell^r}.$$

The collision excess of N is

$$\mathcal{C}_\ell(N) = \sum_{v \in \mathbb{F}_\ell^r} (N(v) - \bar{N})^2.$$

2 The false linear formula

The following expression is a useful failed route:

$$\ell^{-2r} \varphi(P)^2 \sum_{A \in \mathbb{F}_\ell^r \setminus \{0\}} \left(\prod_{q|P} \left[1 + \frac{\ell \mathbf{1}_{A \cdot x_q = 0} - 1}{q - 1} \right] - 1 \right).$$

It is not equal to $\mathcal{C}_\ell(N)$ in general. The failure is structural: Plancherel requires squared complex Fourier coefficients, while the displayed product is a linear real proxy. The factor ℓ^{-2r} and the internal subtraction are also incompatible with the standard finite Fourier normalization.

Example 1 (Minimal sign and normalization failure). *Let $r = 1$, $\ell = 2$, $\mathcal{Q} = \{q\}$, and $x_q = 1$. Then $V(m) = B_q(m) \in \mathbb{F}_2$. Hence*

$$N(1) = |\Gamma_q|, \quad N(0) = q - 1 - |\Gamma_q|.$$

Therefore

$$\mathcal{C}_2(N) = 2 \left(|\Gamma_q| - \frac{q-1}{2} \right)^2 = \frac{(q-1)^2}{2} (1 - 2p_q)^2.$$

The false linear formula gives a different value. For instance, at $q = 3$, $a_1 = 2$, one has $p_q = 1$, so $\mathcal{C}_2(N) = 2$, while the false expression gives a nonmatching linear value. The identity is false as stated.

3 Exact Kummer-Fourier collision variance

Let

$$\zeta_\ell = e^{2\pi i/\ell}.$$

For $A \in \mathbb{F}_\ell^r$, define the finite Fourier coefficient

$$\hat{N}(A) = \sum_{v \in \mathbb{F}_\ell^r} N(v) \zeta_\ell^{A \cdot v}.$$

Theorem 1 (Exact Kummer-Fourier collision variance). *With notation as above,*

$$\mathcal{C}_\ell(N) = \frac{\varphi(P)^2}{\ell^r} \sum_{A \in \mathbb{F}_\ell^r \setminus \{0\}} \prod_{q \in \mathcal{Q}} \left[1 - 2p_q(1 - p_q) \left(1 - \cos \frac{2\pi(A \cdot x_q)}{\ell} \right) \right].$$

Here $A \cdot x_q$ is represented by any integer lift modulo ℓ ; the cosine is well-defined.

Proof. By finite Fourier inversion and Plancherel on $\mathbb{F}_\ell^r$,

$$\mathcal{C}_\ell(N) = \ell^{-r} \sum_{A \neq 0} |\hat{N}(A)|^2.$$

The $A = 0$ term is excluded because subtracting $\bar{N}$ removes the constant component.

For $A \in \mathbb{F}_\ell^r$,

$$\hat{N}(A) = \sum_{m \in (\mathbb{Z}/P\mathbb{Z})^*} \zeta_\ell^{A \cdot V(m)} = \sum_{m \in (\mathbb{Z}/P\mathbb{Z})^*} \prod_{q \in \mathcal{Q}} \zeta_\ell^{B_q(m)(A \cdot x_q)}.$$

Under the uniform measure on $(\mathbb{Z}/P\mathbb{Z})^*$, CRT makes the variables B_q independent, so

$$\hat{N}(A) = \varphi(P) \prod_{q \in \mathcal{Q}} \mathbb{E} \left[\zeta_\ell^{B_q(A \cdot x_q)} \right].$$

For a fixed q , since B_q is Bernoulli with parameter p_q ,

$$\mathbb{E} \left[\zeta_\ell^{B_q(A \cdot x_q)} \right] = (1 - p_q) + p_q \zeta_\ell^{A \cdot x_q}.$$

Thus

$$|\hat{N}(A)|^2 = \varphi(P)^2 \prod_{q \in \mathcal{Q}} \left| (1 - p_q) + p_q \zeta_\ell^{A \cdot x_q} \right|^2.$$

Writing $\theta_q = 2\pi(A \cdot x_q)/\ell$,

$$\left| (1 - p_q) + p_q e^{i\theta_q} \right|^2 = (1 - p_q + p_q \cos \theta_q)^2 + (p_q \sin \theta_q)^2 = 1 - 2p_q(1 - p_q)(1 - \cos \theta_q).$$

Substituting into Plancherel gives the formula. □

Corollary 1 (Binary specialization). *When $\ell = 2$,*

$$\mathcal{C}_2(N) = \frac{\varphi(P)^2}{2^r} \sum_{A \in \mathbb{F}_2^r \setminus \{0\}} \prod_{q \in \mathcal{Q}} [1 - 4p_q(1 - p_q)(1 - \mathbf{1}_{A \cdot x_q = 0})].$$

Proof. For $\ell = 2$, $\cos(\pi t) = 1$ when $t = 0$ in $\mathbb{F}_2$, and $\cos(\pi t) = -1$ when $t = 1$. Substituting in [Theorem 1](#) gives the claim. □

4 Matroid interpretation

Let $M_\ell(\mathcal{Q}; \mathbf{a})$ be the representable Kummer log matroid on ground set $\mathcal{Q}$, represented by the rows $x_q \in \mathbb{F}_\ell^r$. For $A \neq 0$, the condition $A \cdot x_q = 0$ says that the row x_q lies in the hyperplane

$$H_A = \{x \in \mathbb{F}_\ell^r : A \cdot x = 0\}.$$

Thus [Theorem 1](#) is a weighted sum over hyperplanes of the dual arrangement.

Definition 2 (Kummer hyperplane weight). *For $A \neq 0$, define*

$$W_A(\mathcal{Q}) = \prod_{q \in \mathcal{Q}} \left[1 - 2p_q(1 - p_q) \left(1 - \cos \frac{2\pi(A \cdot x_q)}{\ell} \right) \right].$$

Corollary 2 (Weighted hyperplane-arrangement form). *The collision excess is*

$$\mathcal{C}_\ell(N) = \frac{\varphi(P)^2}{\ell^r} \sum_{A \neq 0} W_A(\mathcal{Q}).$$

For $\ell = 2$, W_A depends only on the subset of rows outside the hyperplane H_A :

$$W_A(\mathcal{Q}) = \prod_{q: x_q \notin H_A} [1 - 4p_q(1 - p_q)].$$

Remark 1 (Not an ordinary Tutte evaluation). *The formula is not, in general, a single ordinary Tutte-polynomial evaluation $T_M(x, y)$. The weights are prime-dependent through p_q , and for $\ell > 2$ they depend on the nonzero value of $A \cdot x_q$, not only on whether it vanishes. The honest description is a weighted hyperplane-arrangement sum, equivalent in the binary uniform-weight case to a multivariate Tutte/Potts-style specialization.*

5 Dependency status

- The theorem is exact finite algebra: dependency class E .
- The proof uses only CRT independence and finite Fourier Plancherel.
- No PHNC, BVK, Kummer-BV, GRH, Chebotarev error term, or Erdős–Graham conclusion is used.
- The result strengthens the finite algebraic stack by replacing a false linear product with the correct squared-modulus product.